DIGGIT — Del Amo Environmental Recovery Project

Analytical specification, version 0.1

Research specification
Initial pilot: Del Amo Six target compounds (aromatic hydrocarbon)
Three treatment modules: Sedimentary Processing Therapeutics

Purpose: Establish the scientific, geospatial, biological and computational requirements for a reproducible research application that evaluates contaminant distributions and supports evidence-based remediation decisions at the Del Amo Superfund site. Hydrocarbon contamination is one of the leading urban causes of respiratory disorders and organ dysfunction due to neurochemical contamination. This site presents a unique occasion to develop remediation framework and regulatory protocols associated with industrialization of biological processing to mitigate gross excesses of aromatic hydrocarbon and benzene hyper-accumulation and can serve as a benchmark performance indicator for future urban ecological remediation strategies.

This specification treats DIGGIT as a research and decision-support system, not an autonomous remediation controller. Its recommendations must be traceable to measured data, cited scientific evidence, model assumptions and uncertainty.

The scope is deliberately bounded to:

- Contaminants: benzene, ethylbenzene, naphthalene, phenanthrene, chrysene and benzoapyrene.
- Treatment modules: soil washing, biological treatment and rhizosphere recovery.
- Initial evidence sources: EPA site and analytical records, biological literature, organism and genomic data.
- Primary interface: interactive geospatial workspace with a conversational research agent.
- Primary outputs: contaminant assessment, treatment suitability, research gaps, monitoring recommendations and implementation scenarios.

This follows the governing DIGGIT one-sheet: measured observations, interpolated surfaces and model predictions must remain distinguishable, with provenance and uncertainty preserved.

1. Scientific scope and compound definitions

The six compounds should be represented individually throughout the database and analytical pipeline. Grouping them under BTEX or PAHs is useful for navigation, but insufficient for treatment evaluation.

Compound	Class	Key analytical distinction
Benzene	Monocyclic aromatic VOC	Volatile; groundwater and vapor exposure relevance
Ethylbenzene	Monocyclic aromatic VOC	Volatile; distinct biodegradation and partitioning behavior
Naphthalene	2-ring PAH	More mobile and volatile than higher-ring PAHs
Phenanthrene	3-ring PAH	Intermediate hydrophobicity and sorption
Chrysene	4-ring PAH	Stronger particle association; lower aqueous availability
Benzo[a]pyrene	5-ring PAH	Strong particle association; persistent and toxicologically important

These distinctions determine which measurements, models and treatment-efficacy endpoints are appropriate.

Important analytical rule: a decrease in measured sediment concentration is not automatically evidence of contaminant destruction. DIGGIT must distinguish removal, transfer to another phase, biological transformation and verified mineralization.

2. Treatment-module definitions

Module A — Soil washing

Evaluate physical separation and aqueous partitioning as a means of transferring contaminants from soil or sediment into recoverable liquid or concentrated solids like electrochemical hydrogenation and benzene sorption processes through zeolite and advanced porous polymers articulated for sorption potential of component particulars.
Primary endpoints: residual soil concentrations, recovered contaminant mass, wash-water chemistry, particle-size distribution, secondary waste and mass-balance closure.

Module B — Biological treatment

Evaluate microbial degradation potential using compound-specific literature, functional genomic evidence, site geochemistry and observed microbial community characteristics.
Primary endpoints: parent-compound reduction, transformation products, relevant functional genes, microbial activity, toxicity and evidence of mineralization.

Module C — Rhizosphere recovery
Evaluate plant–microbe–soil interactions for recovery of soil function, vegetation establishment, biological activity and ecological stability.
Primary endpoints: plant establishment, root-zone conditions, microbial community response, soil function, contaminant uptake or transformation, and ecological recovery.

These modules are complementary, not interchangeable. Soil washing is principally a separation process; biological treatment concerns transformation; rhizosphere recovery is an ecological restoration strategy that may also contribute to contaminant management.

3. Site-specific evidence and boundary rules
An important finding from the EPA records affects the data model.
Del Amo is divided into three principal operable units:

- OU-1: Soil and non-aqueous phase liquid (NAPL).
- OU-2: Former waste pits.
- OU-3G: Groundwater, including commingled contamination associated with the nearby Montrose site.

The EPA's contaminant records identify several of the requested compounds at Del Amo, including benzene, ethylbenzene, naphthalene and benzoapyrene, but their recorded media and operable-unit assignments differ. The application must not assume that all six compounds occur in every medium or every OU. cumulis.epa.gov +2

There is also a current-remedy constraint: EPA's 2025 reviews describe the existing soil/NAPL, waste-pit and groundwater remedies as short-term protective, while identifying further long-term actions or monitoring requirements. DIGGIT should therefore evaluate research opportunities alongside the existing remedy, not characterize the site as an uncontrolled or failed cleanup. cumulis.epa.gov +2

Required geographic rule: retain independent OU geometries and a separate representation of the commingled OU-3G groundwater system. Do not merge groundwater observations into Del Amo soil contamination maps.

4. Data schema
The recommended database is PostgreSQL with PostGIS. The schema should be relational at its core, with spatial geometries, time-series observations, document references and versioned analytical outputs.

The key design decision is to separate what was measured, where and when it was measured, what scientific evidence exists, and what DIGGIT infers from those records.

4.1 Core relational entities

Entity	Purpose	Principal fields
project	Research project	project_id, name, description, status
site	Del Amo site	site_id, EPA ID, name, boundary geometry
operable_unit	Separate site management areas	ou_id, site_id, OU designation, geometry
sample_event	Sampling event	event_id, date/time, location, media, depth, method
sample	Physical sample	sample_id, event ID, sample identifier, matrix, custody reference
analyte	Compound dictionary	analyte_id, CAS RN, preferred name, chemical class
analytical_result	Laboratory result	result_id, sample ID, analyte ID, value, units, qualifier, reporting limit
measurement_context	Supporting environmental measurements	pH, moisture, redox, conductivity, temperature, organic carbon
source_record	EPA or external source provenance	source ID, document ID, URL, publisher, publication date
literature_record	Scientific publication	DOI, authors, title, year, abstract, source
organism	Biological candidate	taxon, strain, taxonomy ID, provenance
genomic_record	Genome and sequence evidence	accession, assembly version, annotation source, sequence quality
functional_evidence	Gene/pathway evidence	gene, enzyme, pathway, compound, evidence type
treatment_module	Soil washing, biological, rhizosphere	module ID, definition, applicability
treatment_assessment	Site-specific module assessment	location, compound, module, evidence score, uncertainty
monitoring_record	Follow-up measurements	date, location, endpoint, result
analysis_run	Reproducible computational execution	model version, input snapshot, parameters, timestamp

interpolation_surface	Spatial analytical output	geometry/raster reference, analyte, model, uncertainty
protocol_proposal	Candidate remediation protocol	version, author, assumptions, schedule, status
project_task	Implementation planning	task, dependency, duration, resource and cost estimates
contributor	Community attribution	contributor ID, organization, permissions

4.2 Compound dictionary

Use CAS Registry Numbers as stable chemical identifiers. Compound names alone are insufficient because laboratory datasets may use abbreviations, alternate naming conventions or inconsistent spelling.

Preferred name	CAS RN	Internal code
Benzene	71-43-2	BENZ
Ethylbenzene	100-41-4	EBENZ
Naphthalene	91-20-3	NAPH
Phenanthrene	85-01-8	PHEN
Chrysene	218-01-9	CHRY
Benzo[a]pyrene	50-32-8	BAP

Each analyte record should also store:

- Molecular formula and molecular weight.
- Chemical class and ring count.
- Physical state and volatility classification.
- Water solubility and hydrophobicity reference values.
- Environmental fate reference.
- Relevant analytical methods.
- Toxicological reference links.

Physicochemical properties should be versioned and source-cited rather than hard-coded into analytical functions.

4.3 Analytical result structure

This is one of the most important parts of DIGGIT.

An analytical result must retain the original laboratory result and its interpretation separately.

Field	Example format
sample_id	Foreign key
analyte_id	Foreign key
result_value	Numeric
result_unit	mg/kg, µg/L, µg/kg, etc.
result_qualifier	Detected, nondetect, estimated, rejected
detection_limit	Numeric
reporting_limit	Numeric
analytical_method	EPA or laboratory method
sample_fraction	Whole, dissolved, filtered, etc.
sample_matrix	Soil, sediment, groundwater, soil gas
dry_weight_basis	Boolean / unknown
lab_id	Laboratory reference
qa_status	Validated, qualified, pending
source_record_id	Provenance
import_timestamp	Date/time

Do not convert nondetects to zero. Retain the original qualifier and reporting limit. Statistical models should explicitly handle censored observations. Likewise, do not combine soil concentrations in mg/kg with groundwater concentrations in µg/L into a single concentration surface.

4.4 Spatial and temporal sample structure

Each sample should have:

- Coordinates and coordinate reference system.
- Horizontal accuracy.
- Ground-surface elevation, where available.
- Top and bottom sample depths.
- Sampling date and time.
- Sample matrix and collection method.
- Operable-unit assignment and boundary version.

- Laboratory and chain-of-custody reference.
- QA/QC status.

For groundwater, include monitoring-well identity, screen interval, water-level elevation, hydraulic context and sampling event. For soil gas, retain depth, sampling method and vapor concentration units.

A sample can intersect or be associated with more than one spatial management feature, but the OU assignment must not be inferred solely from a rough map overlay.

4.5 Genomic and biological evidence schema
Biological evidence requires more than a list of organism names.
I recommend five related entities:

- organism_taxon — species, strain, taxonomy identifier.
- genome_assembly — assembly accession, version, completeness and contamination metrics.
- gene_annotation — locus, gene symbol, product, annotation database and confidence.
- pathway_evidence — pathway, substrate, enzyme, products and environmental conditions.
- experimental_evidence — publication, organism, compound, matrix, experimental scale and demonstrated endpoint.

Every organism–compound relationship should have an evidence classification:

Evidence level	Meaning
E0	No relevant evidence located
E1	Predicted gene or pathway annotation
E2	Literature-supported pathway or related compound evidence
E3	Compound-specific laboratory transformation
E4	Compound-specific mineralization or validated degradation
E5	Demonstrated performance in a relevant contaminated environmental matrix
E6	Field-validated performance

These levels are a proposed DIGGIT evidence rubric, not an existing regulatory classification.

A gene annotation must not be treated as proof of active degradation. Genome presence, gene expression, enzyme activity and measured contaminant transformation are separate observations.

5. Biological research and genomic evidence framework

The research database should organize biological evidence by compound, not merely by organism.

5.1 Candidate functional markers

The following are useful starting points for literature and genome searches. They are research markers, not a complete list of genes or a claim that the relevant organisms occur at Del Amo.

Compound	Initial genomic/literature search	Important qualification
Benzene	abcA, aerobic benzene dioxygenase pathways, associated oxygenase genes	Different aerobic and anaerobic pathways; marker specificity matters
Ethylbenzene	todC-related dioxygenase pathways, bssA-associated anaerobic pathways	bssA is relevant to anaerobic alkyl-aromatic metabolism, but must be interpreted in pathway context
Naphthalene	nahAc, nah pathway clusters, ring-hydroxylating dioxygenases	Stronger established pathway literature
Phenanthrene	PAH ring-hydroxylating dioxygenases, phn-associated pathways	Gene families can be substrate- and strain-dependent
Chrysene	High-molecular-weight PAH oxidation literature; bacterial dioxygenases and fungal oxidative enzymes	Compound-specific evidence is more limited; transformation must be verified
Benzo[a]pyrene	PAH oxidation pathways, fungal cytochrome P450, lignin peroxidase, manganese peroxidase and laccase literature	Parent-compound loss is not proof of complete mineralization

A review of diagnostic molecular tools for petroleum-hydrocarbon sites describes markers including abcA, todC, bssA and nahAc, and emphasizes the use of expressed mRNA as evidence of active metabolism rather than relying on DNA detection alone. ngwa.onlinelibrary.wiley.com +1

For PAHs, the literature supports bacterial and fungal transformation pathways, including oxidative enzyme systems. It also identifies the importance of particle sorption and limited bioavailability in soil and sediment. academic.oup.com +2

5.2 Compound-specific biological evidence priorities

Benzene
This should be the strongest site-specific biological research track because EPA documentation already discusses intrinsic benzene biodegradation in the Del Amo/ Montrose groundwater system. DIGGIT should import geochemical indicators such as dissolved oxygen, nitrate, sulfate, ferrous iron, methane, carbon dioxide and oxidation-reduction potential, and connect these to the relevant monitoring wells and sampling dates. semspub.epa.gov +1

Ethylbenzene
Build an independent pathway evidence record. Do not treat benzene biodegradation as evidence of ethylbenzene degradation. Record aerobic and anaerobic pathway evidence separately.

Naphthalene
Prioritize bacterial aromatic-ring oxidation and PAH catabolic pathways. Compare literature on freely available versus sediment-sorbed naphthalene.

Phenanthrene
Prioritize bacterial and fungal transformation literature, particularly the relationship between sediment organic carbon, bioavailability and microbial activity.

Chrysene and benzoapyrene
These require more stringent evidence handling. High molecular weight PAHs can be strongly particle-associated, making biological accessibility a central limitation. DIGGIT should distinguish partial oxidation, transformation-product formation and verified mineralization.

5.3 Cra20: a separate rhizosphere research hypothesis

The uploaded Cra20 paper supports its use as a candidate for plant-growth and drought-resistance research. It reports improved growth and root development in Arabidopsis thaliana, drought-response effects and changes in indigenous rhizosphere bacterial communities. It also reports multiple cold-shock-protein genes in the genome, but that observation is not evidence of sediment freeze–thaw treatment efficacy.

The subsequent complete-genome publication reports xylene and 2,4-D degradation capability. It does not establish degradation of any of the six compounds specified in this project. sciencedirect.com

Therefore, DIGGIT should classify Cra20 as:

- Plant-growth-promoting research candidate.
- Drought-resistance research candidate.
- Rhizosphere-community interaction candidate.
- Xylene/2,4-D degradation evidence available.
- Six target compounds: unverified for Cra20.

This separation is important. Cra20 should not be presented as a demonstrated benzene or PAH remediation organism.

6. Research workflow

The workflow should move from data ingestion to evidence assessment, then to modelling and protocol comparison. Each stage should produce an auditable artifact.

DIGGIT research pipeline

- Stage 1 — EPA data acquisition
  Import site profiles, administrative records, remedial investigation documents, records of decision, five-year reviews, sampling tables and monitoring records.
  Output: source-registered dataset
- Stage 2 — Data harmonization
  Normalize compound identifiers, units, sample matrices, qualifiers, coordinates, dates, depths and OU designations. Preserve original records.
  Output: validated analytical database
- Stage 3 — Compound distribution assessment
  Evaluate concentrations by compound, medium, depth, location, time and OU. Identify spatial and temporal data gaps.
  Output: compound-specific site profiles
- Stage 4 — Biological evidence retrieval
  Search publications, taxonomic records, genome assemblies, gene annotations and functional pathway evidence. Connect each result to the relevant compound and environmental conditions.
  Output: organism–compound evidence matrix
- Stage 5 — Treatment-module assessment
  Compare soil washing, biological treatment and rhizosphere recovery against site conditions, compound properties, scientific evidence and ecological constraints.
  Output: module suitability assessments

- Stage 6 — Spatial and predictive modelling
  Develop appropriate interpolation surfaces, temporal trend models and climate-aware scenario analyses. Quantify uncertainty and validate predictions.
  Output: analytical maps and forecasts
- Stage 7 — Protocol comparison
  Generate alternative research and remediation scenarios with cost, environmental risk, scheduling, monitoring and evidence requirements.
  Output: comparative decision report
- Stage 8 — Community review and monitoring
  Record researcher feedback, field observations, implementation status, updated laboratory results and changes in evidence.
  Output: versioned research history

6.1 EPA data ingestion requirements
EPA's Del Amo site profile and document repository should be the initial source of truth for site-specific records.

The ingestion process should prioritize:

- EPA Superfund site profile and contaminant records.
- Operable-unit boundaries and site documents.
- OU-1 soil/NAPL remedial investigation and feasibility study.
- OU-2 waste-pit investigations and monitoring records.
- OU-3G groundwater investigations and annual monitoring records.
- 2025 Five-Year Review reports and associated appendices.
- Historical and current analytical tables.

EPA maintains separate document collections for OU-1, OU-2, OU-3 and five-year reviews. These should be ingested as distinct document groups, with source metadata and extraction status retained. cumulis.epa.gov +1

Critical limitation: EPA's public contaminant-of-concern table is not a complete analytical dataset. It identifies chemicals and media associated with remedy decisions; it does not replace the underlying sample-level measurements. DIGGIT must retrieve the actual analytical tables before generating quantitative contaminant maps.

6.2 Data quality and validation
Every imported result should pass the following checks:

- Valid compound identity and CAS number.

- Valid units and matrix.
- Coordinate reference system and spatial validity.
- Sample depth and date consistency.
- Laboratory qualifier and detection-limit preservation.
- Duplicate and blank identification.
- Outlier flags without automatic deletion.
- OU assignment review.
- Original source-document linkage.

A result can be valid but qualified, or scientifically useful but unsuitable for a particular model. Those states must not be conflated.

6.3 Spatial interpolation strategy

Interpolation should be performed independently for each compound and compatible sample matrix.

For example:

- Soil benzene: soil samples, consistent depth interval and appropriate OU.
- Groundwater benzene: monitoring wells and groundwater sampling events.
- Soil-gas benzene: soil-gas observations, not soil or groundwater values.
- PAHs: individual compound surfaces, with depth and particle characteristics retained.

Candidate methods:

Method	Application	DIGGIT requirement
IDW	Initial exploratory	Clearly labelled exploratory
Ordinary kriging	Spatially structured	Variogram assessment and cross-validation
Universal kriging	Spatial trend plus	Justified trend model
Bayesian spatial	Sparse observations	Explicit prior and posterior diagnostics
Temporal/	Repeated sampling	Account for monitoring history and changing

Do not generate a visually continuous contamination surface merely because a GIS can interpolate between points. Where the data are sparse or unsuitable, display sample points, coverage gaps and uncertainty instead.

6.4 Biological suitability workflow

For each compound, the biological evidence engine should ask:

- Has the organism been tested against this exact compound?
- Was the compound transformed, degraded or mineralized?
- What was the experimental matrix?
- Were relevant genes detected, expressed or functionally validated?
- What redox, moisture, temperature and nutrient conditions were reported?
- Were transformation products and toxicity evaluated?
- Was the evidence obtained in pure culture, mixed community, contaminated soil or a field environment?
- Does the environmental context resemble Del Amo?

The resulting assessment should distinguish a plausible biological mechanism from a demonstrated site-relevant treatment.

7. Analytical outputs
The first demonstrator should generate six classes of analytical output.

7.1 Compound distribution profiles
For every compound:

- Concentration distribution by medium.
- Concentration distribution by OU.
- Depth profile.
- Sampling chronology.
- Detection frequency and nondetect proportion.
- Maximum, median, mean and relevant quantiles.
- Spatial coverage and uncertainty.
- Available comparison with cleanup criteria or site-specific decision thresholds.

Thresholds should be stored with their legal or scientific source, media, land-use assumptions, effective date and applicability. A generic soil screening value must not be treated as an automatically applicable cleanup target.

7.2 Contaminant and geochemistry maps
The map interface should support:

- Individual compound layers.
- Combined compound-group views.
- Sampling locations and depths.
- Soil, groundwater and soil-gas separation.

- pH, redox, dissolved oxygen and conductivity.
- Organic carbon and moisture.
- Measured versus interpolated versus predicted layer selection.
- Confidence or uncertainty overlays.
- Time-slider comparisons.

7.3 Biological evidence matrix

A searchable matrix should compare compounds and candidate organisms.

Compound	Bacterial evidence	Fungal evidence	Cra20 evidence
Benzene	Established pathway literature	Additional literature review	Unverified
Ethylbenzene	Established pathway literature	Additional literature review	Unverified
Naphthalene	Strong pathway literature	Established transformation literature	Unverified
Phenanthrene	Established transformation literature	Established transformation literature	Unverified
Chrysene	High-MW PAH evidence; strain-specific validation needed	Transformation literature	Unverified
Benzo[a]pyrene	Transformation evidence; mineralization needs verification	Oxidative enzyme evidence	Unverified

This is a literature-level screening matrix, not evidence that any particular organism is present or active at Del Amo. pubmed.ncbi.nlm.nih.gov +2

7.4 Treatment suitability matrix

DIGGIT should produce a compound-specific assessment for each treatment module.

Compound	Soil washing	Biological treatment	Rhizosphere recovery
Benzene	Evaluate volatilization, aqueous transfer and containment	Strong research basis; site-specific biodegradation context	Supporting ecological recovery, not presumed primary removal
Ethylbenzene	Evaluate phase transfer and recovered liquid	Compound-specific pathway assessment	Supporting recovery
Naphthalene	Evaluate washing and recovery potential	Strong research basis	Potential plant–microbe contribution
Phenanthrene	Evaluate sorption and extraction limitations	Strong research basis	Relevant rhizosphere research

Chrysene	Evaluate particle-associated fraction and separation	Higher uncertainty; verify products	Long-term ecological recovery research
Benzo[a]pyrene	Evaluate strongly sorbed fraction and secondary waste	Higher uncertainty; transformation versus mineralization distinction	Long-term recovery; uptake and exposure constraints

For soil washing, the literature indicates that extraction-agent selection and soil characteristics substantially influence PAH removal. It also emphasizes that the washing process produces concentrated effluents requiring their own treatment and recovery strategy. pubmed.ncbi.nlm.nih.gov +1

7.5 Mass-balance and treatment-performance report
Every proposed treatment assessment should use a mass-balance framework:

$$M_{\mathrm{initial}} = M_{\mathrm{residual\ soil}}+M_{\mathrm{water}}+M_{\mathrm{captured}}+M_{\mathrm{verified\ products}}+M_{\mathrm{unaccounted}}$$

DIGGIT should calculate:

- Parent-compound mass removed from the original matrix.
- Parent-compound mass recovered in secondary streams.
- Verified transformation products.
- Evidence of mineralization where available.
- Unaccounted mass and analytical uncertainty.
- Residual toxicity and ecological effects.

A high unaccounted fraction should prevent the application from assigning a high treatment-success rating.

7.6 Remediation planning and cost analysis
The planning output should contain:

- Candidate treatment sequence.
- Estimated project duration.

- Resource and material requirements.
- Cost ranges with assumptions.
- Sampling and monitoring schedule.
- Weather-sensitive constraints.
- Environmental risk flags.
- Required regulatory and technical review.
- Evidence gaps and unresolved questions.

The platform should not invent cost estimates where no reliable pricing inputs exist. Unknown costs should be identified as unknown rather than represented by false precision.

8. Implementation requirements

8.1 Recommended technology stack

Component	Recommended technology	Responsibility
Web interface	HTML5, JavaScript, React	Mobile and desktop application
Geospatial GUI	MapLibre GL JS	Interactive mapping and layer controls
Spatial database	PostgreSQL + PostGIS	Sites, samples, geometries and spatial queries
Backend	Python + FastAPI	API, authentication and analysis orchestration
Scientific computing	SciPy, GeoPandas, scikit-learn	Geostatistics and statistical analysis
Raster processing	GDAL, Rasterio	Satellite and terrain data
Biological data	NCBI Datasets, NCBI Taxonomy, KEGG where licensed, UniProt	Genome and functional annotation retrieval
Literature	Crossref, PubMed, Europe PMC	Publication metadata and literature discovery
Document storage	S3-compatible object storage	EPA documents, laboratory reports and contributed files
Authentication	OAuth 2.0 / OpenID Connect	Researcher identities and access control

8.2 Required application modules

Organize the application into seven principal workspaces.

1. Geospatial Research

Central map, site boundaries, OU navigation, sampling locations, contaminant layers and spatial interpolation.

2. Chemical Analysis
Six-compound database, laboratory results, geochemical profiles, phase distribution and mass balance.

3. Biological Intelligence
Organism records, genomes, functional genes, microbial pathways and compound-specific literature.

4. Rhizosphere and Ecological Recovery
Plant suitability, root-zone conditions, microbial interactions and ecological recovery indicators.

5. Predictive Analytics
Spatial interpolation, time-series analysis, climate relationships and uncertainty assessment.

6. Remediation Planning
Protocol comparisons, cost modelling, schedules, resource constraints and monitoring dependencies.

7. Community Research
Research contributions, peer review, project attribution, shared datasets and version histories.

8.3 Intelligence engine requirements
The DIGGIT Research and Remediation Agent should have access to controlled analytical functions, not unrestricted authority over the underlying scientific data.
Its capabilities should include:

- Natural-language research queries.
- Compound-specific literature retrieval.
- Geospatial query construction.
- Sample and measurement aggregation.
- Statistical analysis requests.
- Biological evidence comparison.

- Scenario generation.
- Report compilation.
- Explanation of uncertainty and missing data.

Every AI-generated conclusion should include:

- The underlying source records.
- The analytical method used.
- The relevant model version.
- The assumptions and limitations.
- An evidence-strength designation.
- A reproducible analysis-run identifier.

The AI should never create missing sample measurements, silently fill nondetects, or represent a predicted result as a laboratory observation.

8.4 Security and research governance

The application needs:

- Role-based access control.
- Researcher attribution and licensing metadata.
- Version-controlled datasets and protocols.
- Immutable source-document references.
- Audit trails for changes to analytical records.
- Explicit distinction between public and restricted information.
- Consent and review procedures for contributed field data.
- Scientific and regulatory review before implementation of proposed interventions.

Biological introductions, amendments and treatment changes must remain subject to qualified ecological, biosafety and regulatory assessment. DIGGIT can evaluate evidence and prepare research proposals, but it should not autonomously authorize field interventions.

9. Initial analytical outputs and acceptance criteria

The first release should be considered successful when it can demonstrate the following.

Deliverable	Acceptance criterion
EPA site database	Source documents and available analytical records imported with provenance
OU geography	OU-1, OU-2 and OU-3G remain independently identifiable

Six-compound registry	All six compounds resolve through CAS identifiers
Sample database	Original units, qualifiers, limits and spatial metadata retained
Contaminant maps	Measurements and interpolations visibly distinguished
Biological evidence library	Literature and genomic evidence linked to compounds and organisms
Cra20 profile	Plant-support evidence separated from unverified hydrocarbon degradation
Treatment comparison	All three modules evaluated independently
Analytical reports	Reproducible calculations and traceable citations
Planning engine	Cost, schedule, environmental constraints and uncertainty represented
Community workspace	Contributor attribution and version history supported

9.1 Recommended release sequence

Release 1 — Data foundation
First priority
EPA records, six-compound registry, sample schema, OU boundaries and map interface.

Release 2 — Scientific intelligence
Biological literature, genome annotations, compound-specific evidence grading and Cra20 research profile.

Release 3 — Analytical modelling
Interpolation, geochemistry, temporal analysis, treatment suitability and uncertainty.

Release 4 — Planning and community
Cost and scheduling comparisons, monitoring plans, collaborative research and AI-assisted reporting.

Let's evade starting with sophisticated machine learning. The first demonstrator needs reliable source data, a correct relational schema and reproducible analytical functions. Predictive analytics becomes much more valuable once those foundations are established.

10. Principal scientific questions for the Del Amo pilot

These should become the initial research-question registry inside DIGGIT.

Research question	Primary data required
Where are the six compounds documented, and in which media and OUs?	EPA analytical records
How do concentrations vary with depth and time?	Sampling history and laboratory data
How do sediment properties affect PAH partitioning and availability?	Organic carbon, texture and chemical measurements
What biological pathways have evidence for each compound?	Literature, genomes and functional annotations
Which environmental conditions are associated with benzene biodegradation?	Groundwater geochemistry and biological indicators
Can Cra20 contribute to rhizosphere recovery under site-relevant conditions?	Plant, soil and microbial evidence
How do soil washing, biological treatment and rhizosphere recovery compare?	Compound-specific treatment evidence and site characteristics
What environmental and operational constraints affect treatment schedules?	Climate, hydrology, cost and project data

11. Important initial scientific references

These are useful starting points for the biological and treatment evidence library:

- Bacterial degradation of aromatic compounds — aromatic compound catabolic pathways.
- Microbial biodegradation of polyaromatic hydrocarbons — bacterial and fungal PAH degradation.
- Application of Diagnostic Tools to Evaluate Remediation Performance — molecular indicators of hydrocarbon biodegradation.
- Extraction agents for the removal of PAHs from soil — soil-washing extraction-agent review.
- Removal of hydrophobic organic pollutants from soil washing solutions — secondary liquid-stream treatment.
- Complete genome sequence of Sphingomonas sp. Cra20 — Cra20 genomic research.
- 2025 Del Amo–Montrose groundwater Five-Year Review — groundwater monitoring and intrinsic biodegradation evidence.
- EPA Del Amo reports and documents — primary site document repository.

12. Final development definition

I would place the following statement at the beginning of the technical specification:
DIGGIT Del Amo Pilot 001 is a geospatial environmental research and decision-support application designed to integrate EPA site records, compound-specific analytical

measurements, biological literature, microbial genomic data and ecological recovery indicators. Its purpose is to characterize the distribution and environmental behavior of six target aromatic contaminants, evaluate soil washing, biological treatment and rhizosphere recovery as independent and potentially complementary remediation modules, and support evidence-based research and implementation planning through geospatial interpolation, scientific analysis and predictive modelling.

The six compounds and three treatment modules form a sufficiently bounded first research demonstrator, while the underlying architecture remains reusable for other contaminated sites.

Atmospheric Remediation Within Reach

Immediate development, an artifact of the PostgreSQL/PostGIS database schema and its initial EPA ingestion specification. That gives us a concrete foundation for the application rather than another conceptual presentation. This document establishes what the first working system must actually contain. The Del Amo aromatic hydrocarbon discharge EPA Superfund site serves as a most appropriate model for a beta version pilot remediation pipeline candidate in that it presents a unique opportunity to develop applications for mitigating pollutants accrued of rubber processing and arrives with a surplus of relatively current and consistent data for calculating data trends unmitigated and surveying and analysis data for scheduling sedimentary remediation processes and development processes and protocols to mitigate hazards associated with with groundwater contamination and aromatic hydrocarbon containment processes such as temporary enclosed worksite containment systems and activated metal groundwater filtration sedimentary filtration layering techniques. By providing a community based, efficiency optimized dataset aggregate solution to ecological dysfunction, the Digital Interface for Graphing Geochemical Interpolation Treatment, DIGGIT, will be a fundamental tool for institutions, organizations, and landholders in developing remediation therapy solutions to geological contamination issues. With great care and concentrated effort, atmospheric remediation is within reach and environmental issues become geocivil solutions.

7 November, 2026

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